a one-way relationship in either direction. It's important to ensure a mutual relationship. In the data mesh context, data contracts and data sharing agreements are there to help accomplish just that.

Data contracts and data-sharing agreements are still largely topics of ongoing discussion, but they seem to emerge as one possible tool to foster this mutual consumer- producer relationship. We discuss these options next and then outline how to get started with them.

5.7.1 Data contracts and sharing agreements

Piethein Strengholt describes *data contracts* as delivery or service contracts (see "Data Contracts—Ensure Robustness in Your Data Mesh Architecture," http://mng.bz/gwnE). If we look at the preceding Cast data product, a data contract could include information like this:

- This is a production-level dataset.
- The dataset is checked for broken records and does not contain test data.
- The interface delivers the data within one second inside the company network.
- We guarantee 95% uptime.
- The roadmap includes a couple of new datasets to be released over the next few months.
- The data product is currently in version 1.0.10.
- We use semantic versioning. That means a bump to 1.0.*X* means bug fixes, and a bump to 1.*X*.0 means we added data but did not break or remove anything. Both of these types of changes are downward compatible.
- The schema of the data and any additional metadata.

A *data-sharing agreement* extends on this data contract. While the data contract is still heavy on the data-producer side, the data-sharing agreement specifically targets consumers. It is there to answer the questions of how a dataset is planned to be used and the needed level of security. Data-sharing agreements, therefore, involve the collaboration of producers and